



WASHINGTON, D.C.
May 26-29, 2026
Pre-Conference: May 25-26, 2026



Mastering Power BI Visual Development: From Data to Storytelling

Andrew Sommer

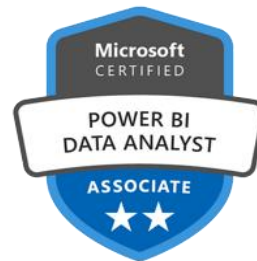
Chief Data Officer | **MISSOURI S&T**



Brianna Reddick



Associate Director of Analytics | **MISSOURI S&T**



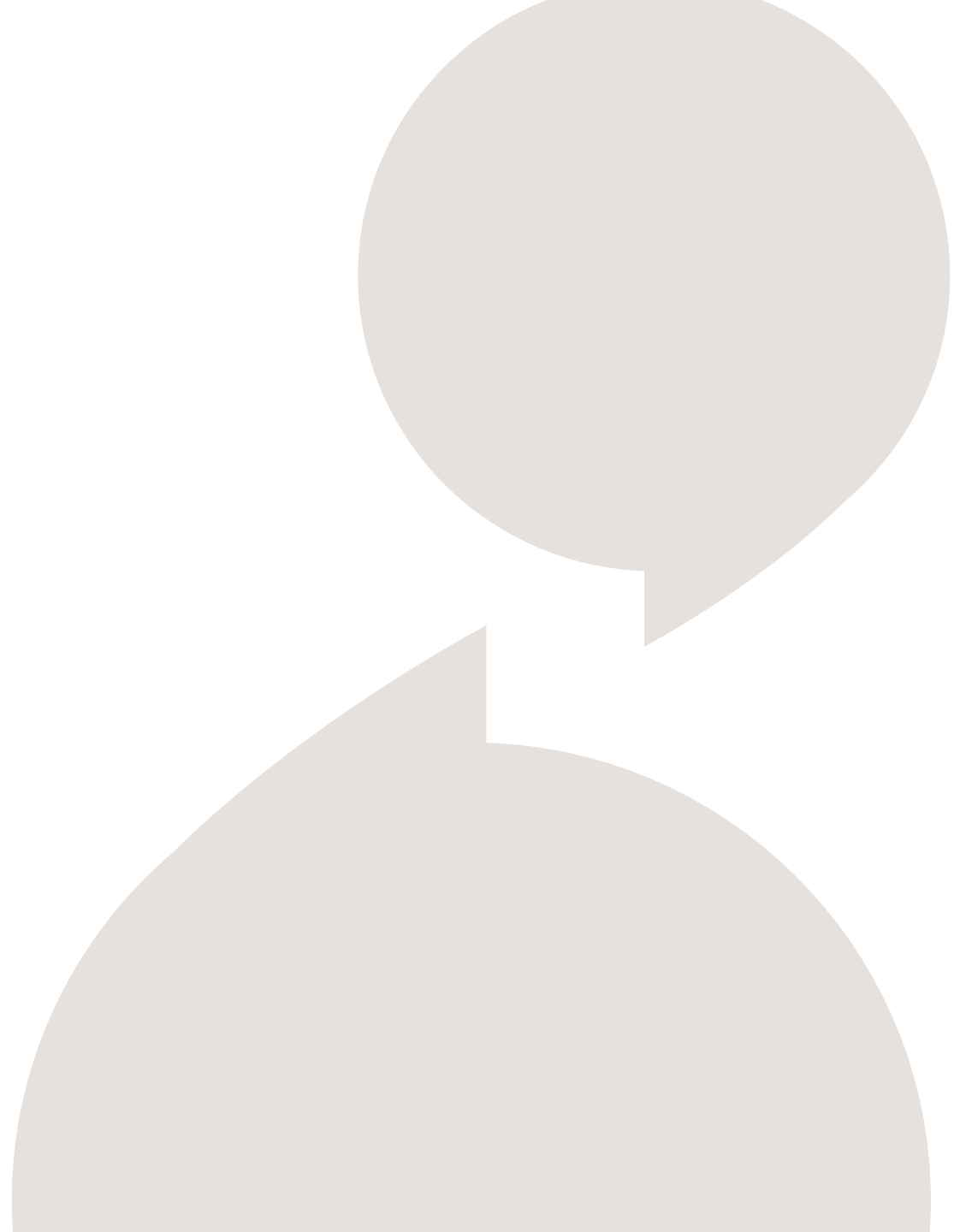
Santhana Lakshmi Ponnurasan

Power BI 2025 & 2026 World Championship Finalist

Data Analyst | MISSOURI S&T



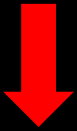
Intro + Overview





Microsoft Fabric

The unified data platform for AI transformation



Data
Factory



Analytics



Databases



Real-Time
Intelligence



IQ



Power BI

Fabric Platform



Copilot



OneLake



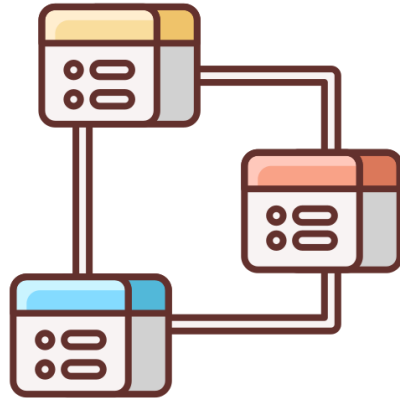
Governance

The Power BI Workflow



1. Connect to Data

Single source
Multiple sources
Cloud & on-premises



2. Shape the Data

Clean & transform
Apply queries
Build models



3. Create Dashboards & Reports

Charts & graphs
Interactive elements
Compelling storytelling

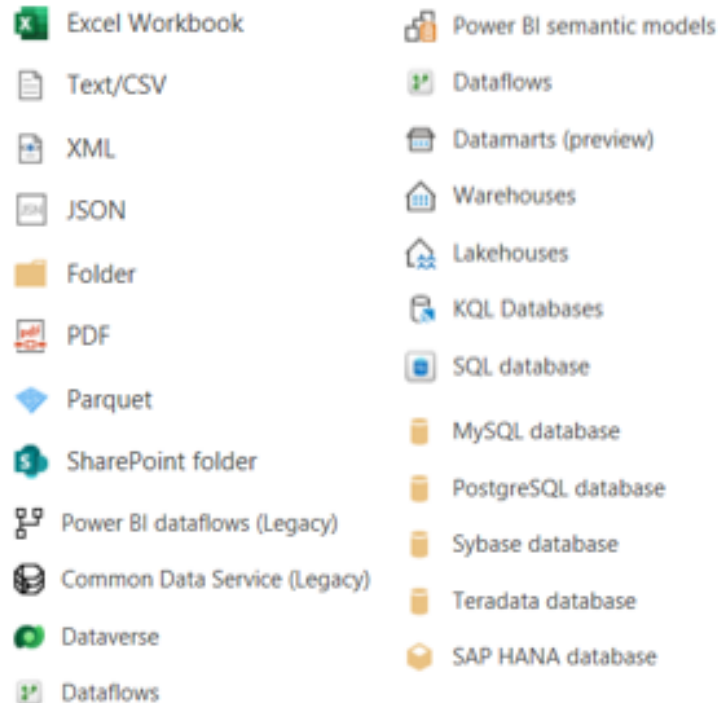


4. Share Reports

Share .pbix files directly
Upload to Power BI Service

Data Connectors

Data Sources to ingest or pull data from



Connectivity Modes

Methods of establishing connection

1. Import
2. Direct Query
3. Live Connect
4. Mixed/Composite
5. Direct Lake

Data Source Types

File

-  Excel Workbook
-  Text/CSV
-  XML
-  JSON
-  Folder
-  PDF
-  Parquet
-  SharePoint folder

Microsoft Fabric

-  Power BI semantic models
-  Dataflows
-  Datamarts (preview)
-  Warehouses
-  Lakehouses
-  KQL Databases
-  SQL database

Azure

-  Azure SQL database
-  Azure Synapse Analytics SQL
-  Azure Analysis Services database
-  Azure Database for PostgreSQL
-  Azure Blob Storage
-  Azure Table Storage
-  Azure Cosmos DB v1
-  Azure Data Explorer (Kusto)
-  Azure Data Lake Storage Gen2
-  Azure HDInsight (HDFS)
-  Azure HDInsight Spark
-  HDInsight Interactive Query
-  Azure Cost Management
-  Azure Device Registry (ADR) (Beta)
-  Azure Resource Graph
-  Azure Cosmos DB v2

Database

-  SQL Server database
-  Access database
-  SQL Server Analysis Services database
-  Oracle database
-  IBM Db2 database
-  IBM Informix database (Beta)
-  IBM Netezza
-  MySQL database
-  PostgreSQL database
-  Sybase database
-  Teradata database
-  SAP HANA database
-  SAP Business Warehouse Application Server
-  SAP Business Warehouse Message Server
-  Amazon Redshift
-  Impala

Online Services

-  SharePoint Online List
-  Microsoft Exchange Online
-  Dynamics 365 Online (legacy)
-  Dynamics 365 (Dataverse)
-  Dynamics NAV
-  Dynamics 365 Business Central
-  Dynamics 365 Business Central (on-premises)
-  Azure DevOps (Boards only)
-  Azure DevOps Server (Boards only)
-  Salesforce Objects
-  Salesforce Reports
-  Google Analytics
-  Adobe Analytics
-  appFigures (Beta)
-  Data.World - Get Dataset (Beta)
-  GitHub (Beta)

Power Platform

-  Power BI dataflows (Legacy)
-  Common Data Service (Legacy)
-  Dataverse
-  Dataflows

Other

-  Web
-  SharePoint list
-  OData Feed
-  Active Directory
-  Microsoft Exchange
-  Hadoop File (HDFS)
-  Spark
-  Hive LLAP
-  R script
-  Python script
-  ODBC

Connection Modes



Data



Model



Visuals

Import

Data is imported into Power BI and stored in the **in-memory** model (fast, but file size limits apply).

Data Source



Direct Query

Data remains in the **source** system; queries are sent live whenever visuals are refreshed (no data stored in PBIX and is slow).



Live Connection

Similar to DirectQuery but used with Analysis Services/ Power BI datasets; **no model is stored in the PBIX**.



Mixed/Composite

A **mix** of Import & DirectQuery in the same model, giving flexibility & balance between performance and real-time.



Direct Lake

Handles large data to be quickly loaded into memory from **delta tables** available in the OneLake. (Needs **Fabric Capacity**)



Stored within
Power BI



Game Plan

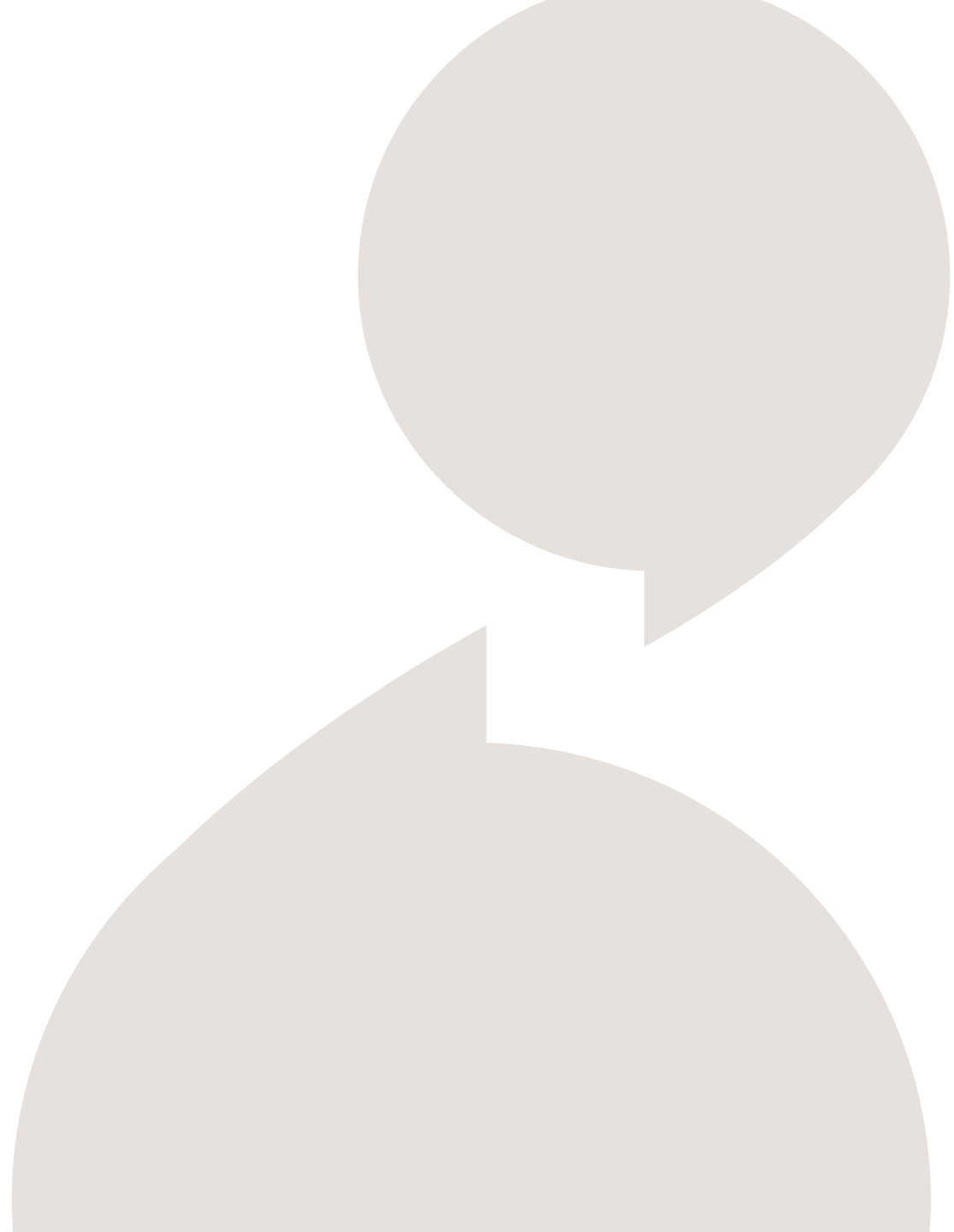


1. Data Modeling
2. Why Storytelling?
3. Why Dashboards Fail?
4. What makes a “good” dashboard?
5. Designing Before Building
6. KISS Principle for Designing
7. Hands-On Dashboard building

Download resources from

<https://tinyurl.com/AIR2026PowerBI>

Foundation First: Data Modeling for Better Reports



WHY A STRONG DATA MODEL MATTERS



TRUSTED INSIGHTS

Accurate data leads to confident, data-driven decisions.



BETTER PERFORMANCE

Optimized models deliver faster reports and smoother experiences.



EASY MAINTENANCE

Well-structured models are easier to update and scale.



BUSINESS ALIGNMENT

A clear model ensures everyone speaks the same data language.



FUTURE READY

Flexible models adapt as your business grows.

“Data should be transformed as far upstream as possible, and as far downstream as necessary.”

-Roche's Maxim of Data Transformation

BEST PRACTICES



Start with business requirements and key questions.



Use a star schema (facts & dimensions).



Define primary keys and relationships clearly.



Filter early, load only what you need.



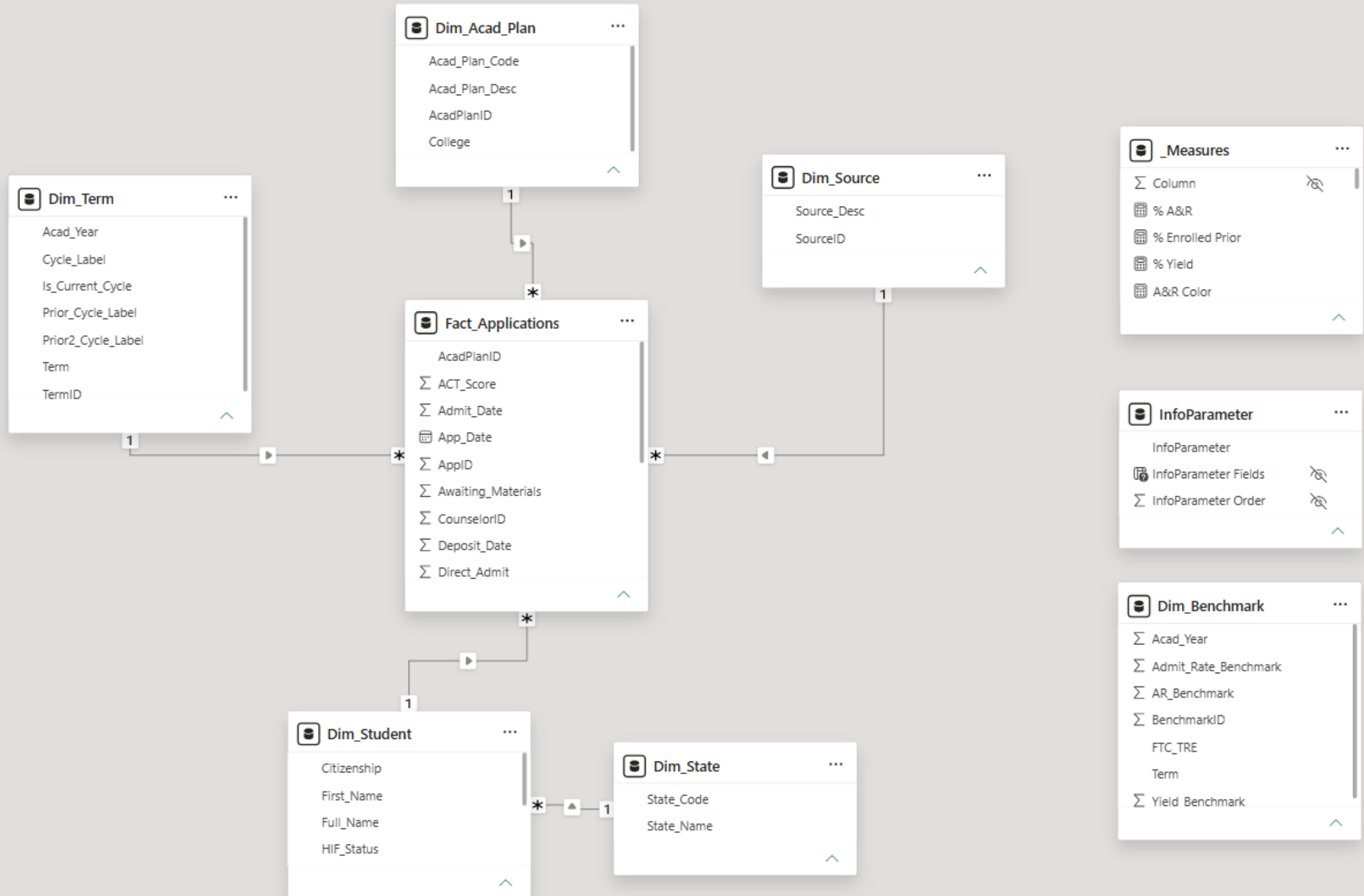
Use proper data types and formats.



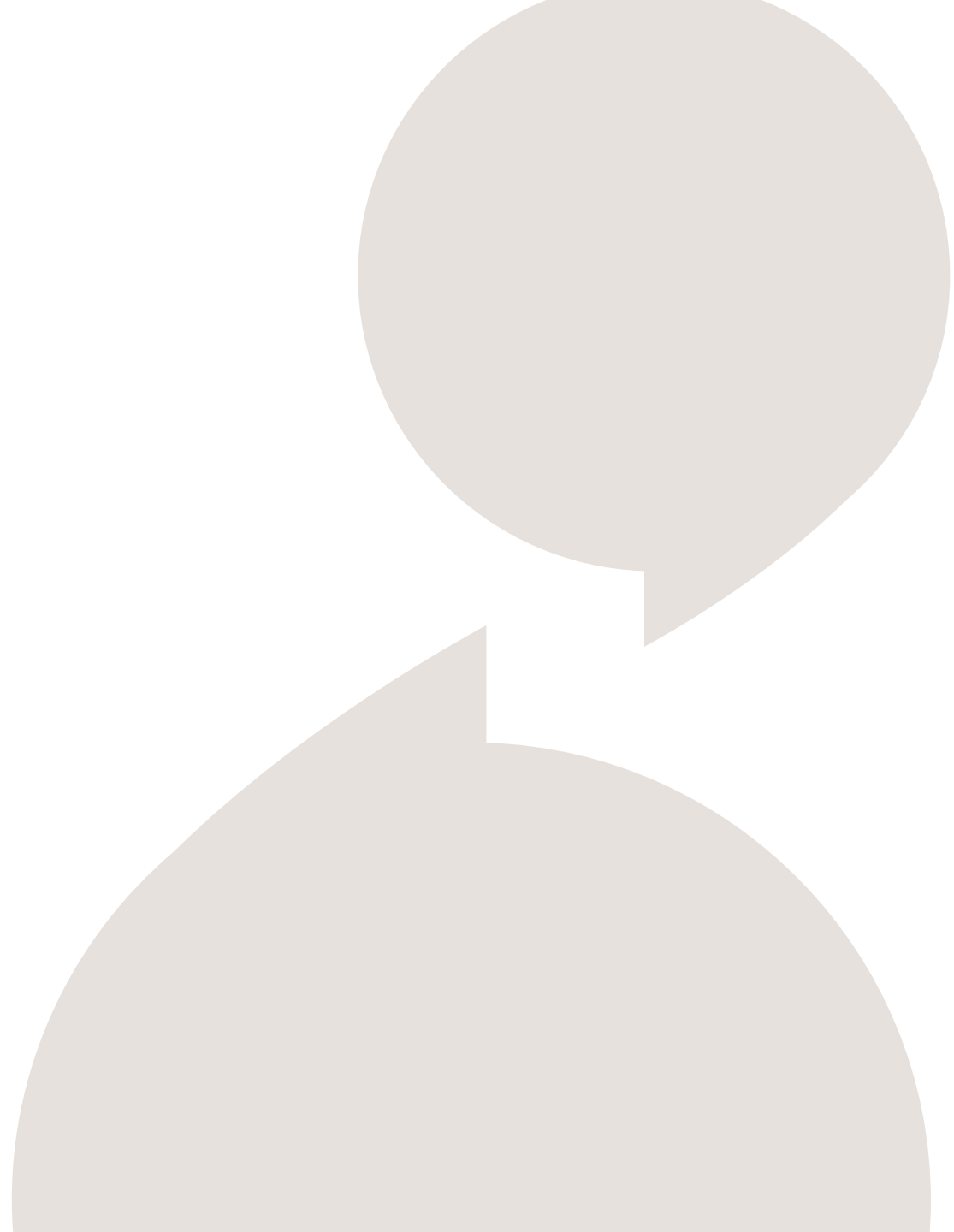
Validate data regularly for accuracy.



Document your model for clarity and reuse.



The “Why” Behind Visual Development?



The Challenges



**You have
tons of data**



**Your audience has
limited time**



**Attention is
fragmented**



**Stakeholders have
different priorities**



**Audiences forget
statistics quickly**



The Opportunities



You have tons of data



➔ **Opportunity:**

Transform overwhelming reports into clear, memorable narratives that highlight what matters most.



Your audience has limited time



➔ **Opportunity:**

Use concise stories to communicate insights faster and keep attention longer than spreadsheets alone.



Attention is fragmented



➔ **Opportunity:**

Use compelling stories to cut through distractions and keep audiences focused on key messages.



Stakeholders have different priorities



➔ **Opportunity:**

Tailor stories for leaders, faculty, students, alumni, and donors so each group sees what matters to them.



Audiences forget statistics quickly



➔ **Opportunity:**

Turn facts into narratives people remember and repeat.



Why Storytelling Wins?

 DATA +  STORY =  IMPACT



Builds Connection

Stories create emotional connections that inspire and resonate.



Strengthens Identity

Stories communicate who we are, what we value, and where we're going.



Enhances Engagement

Compelling stories capture attention and deepen engagement.



Drives Impact

Stories highlight outcomes and demonstrate the real-world impact of education.



Inspires Action

Great stories motivate audiences to support, enroll, and advocate.





WASHINGTON, D.C.
May 26-29, 2026
Pre-Conference: May 25-26, 2026



Making Higher Ed Data Speak – Storytelling Frameworks That Drive Impact

**May 27, 2026
1:45pm – 2.30pm
Magnolia Room**



Why Dashboards Fail

1



TOO MUCH
INFORMATION

2



NO CLEAR STORY
OR PRIORITY

3



COMPARISONS
LACK CONTEXT

4



INCONSISTENT
DESIGN

5



BUILT FOR ANALYSTS,
NOT DECISION-MAKERS

6

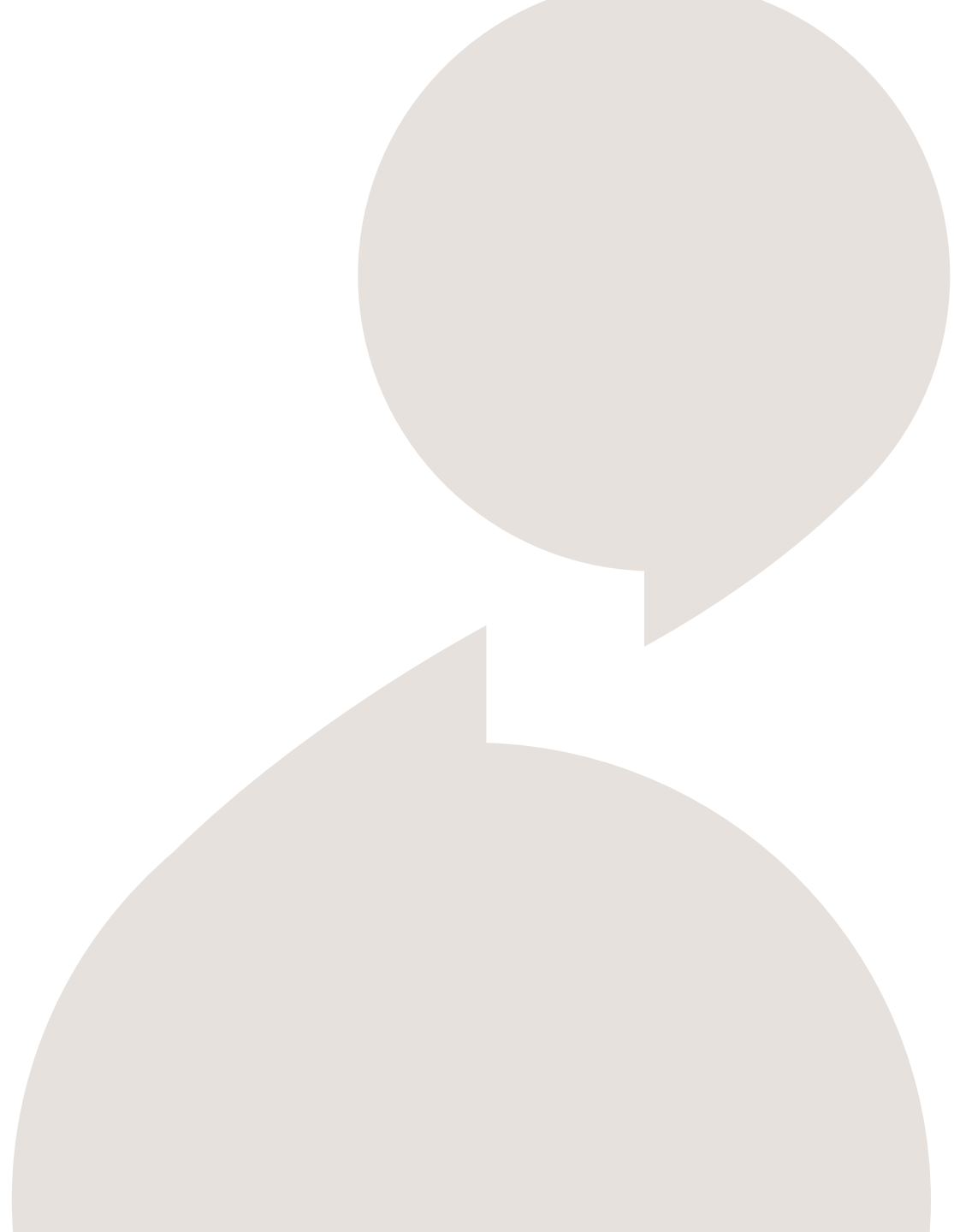


NO CLEAR
ACTION



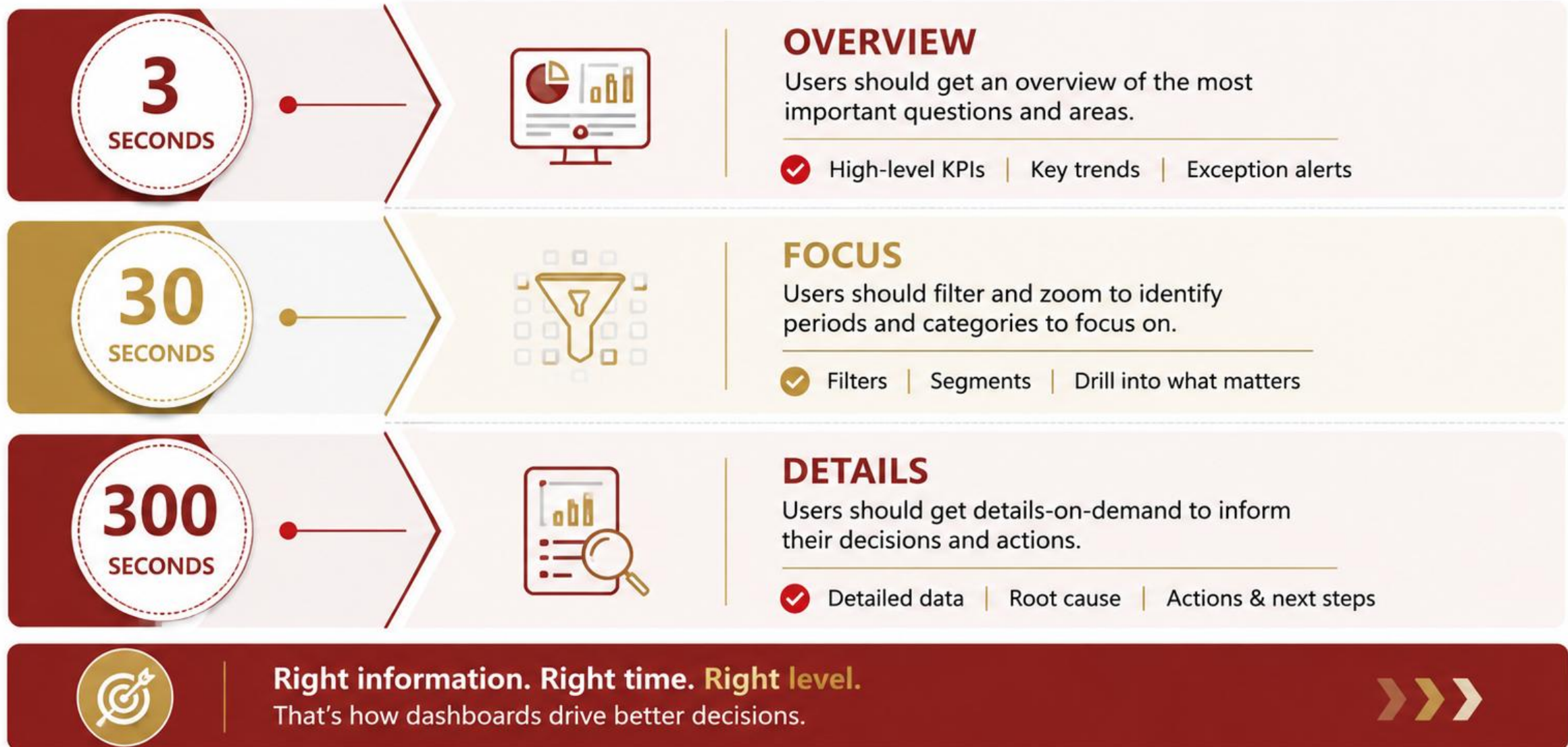
A dashboard is successful only if someone can **understand** and **act** on it quickly.

**What makes a “good”
dashboard?**

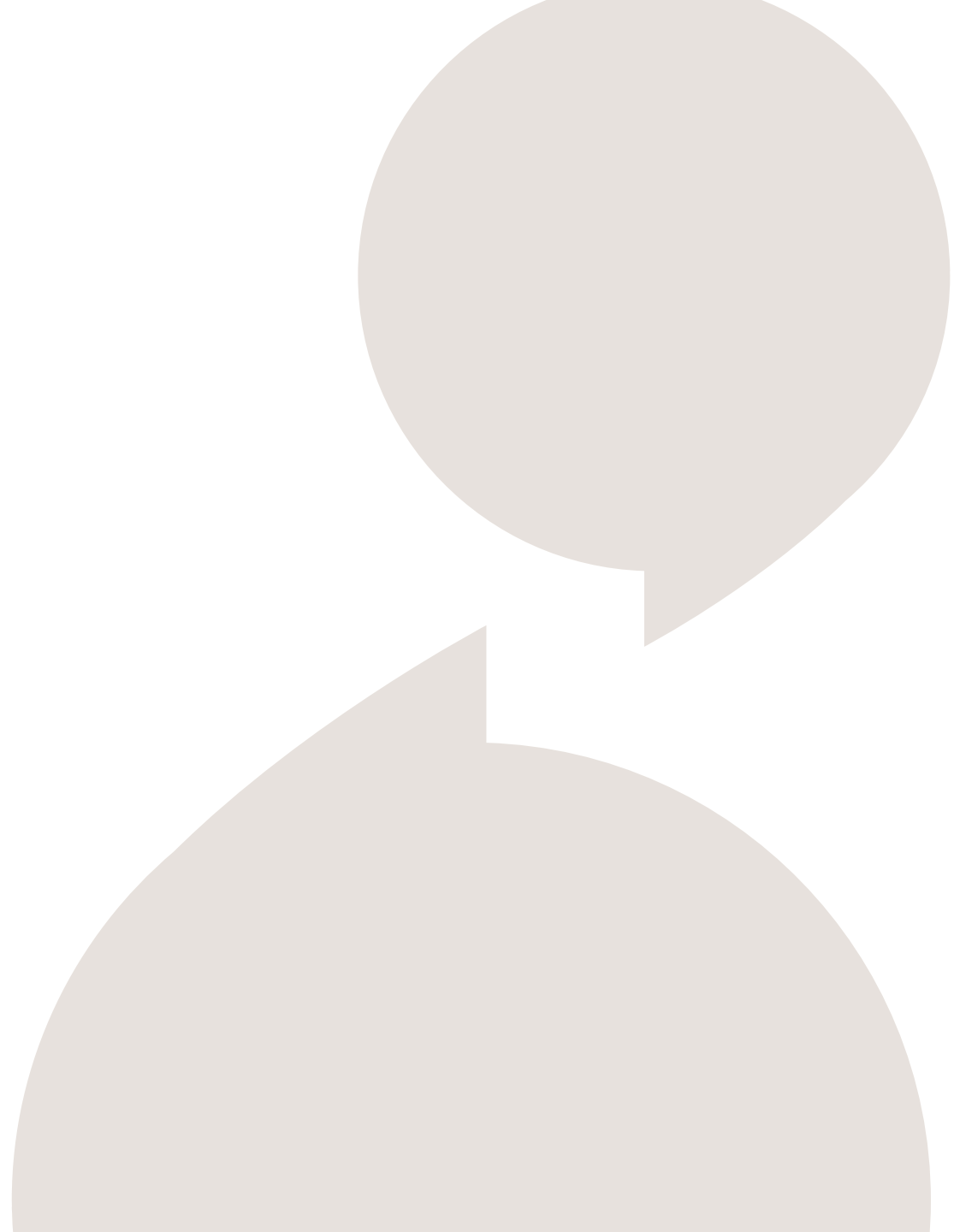


The 3-30-300 Rule

Design dashboards that meet users where they are in their decision journey.



**What are we building
today?**



FTC

TRE

Fall

Spring

☐ College

☐ Source

☒ Acad Plan

☐ Residency

☐ State

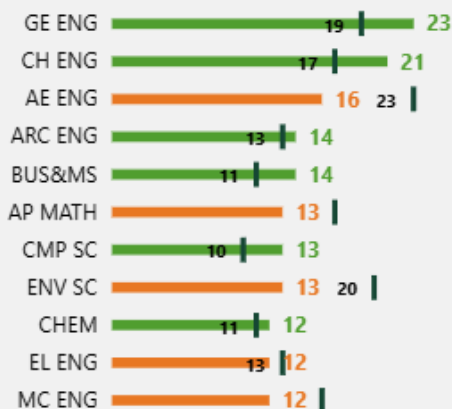
☐ Sex

☐ Citizenship


Apps #

254

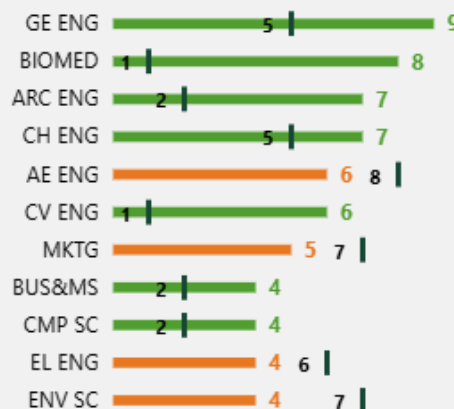
Prior Cycle 290 ▼ -36 (-12.4%)



Admits #

90

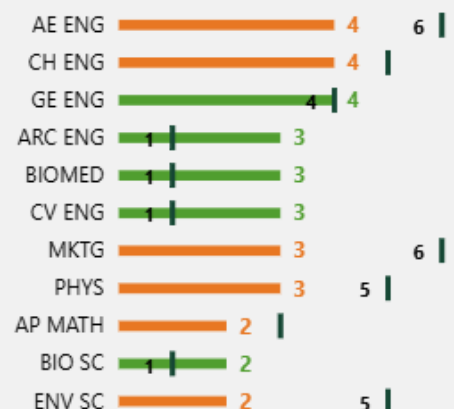
Prior Cycle 81 ▲ 9 (11.1%)



Net Deposits #

43

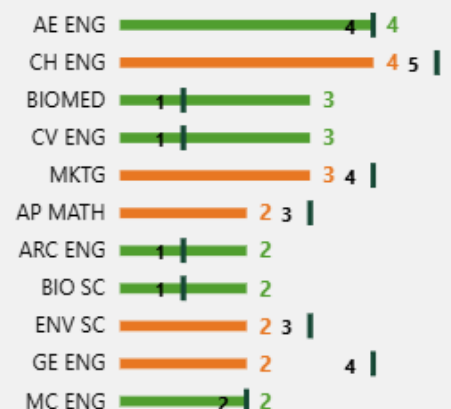
Prior Cycle 69 ▼ -26 (-37.7%)



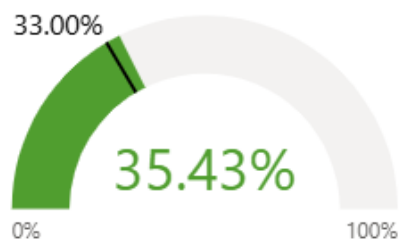
Enrolled #

38

Prior Cycle 55 ▼ -17 (-30.9%)

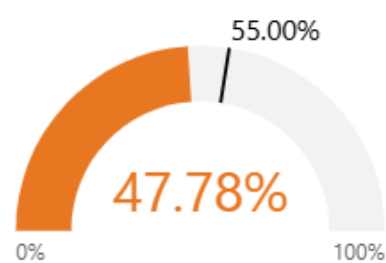


Admit % vs Admit % Benchmark



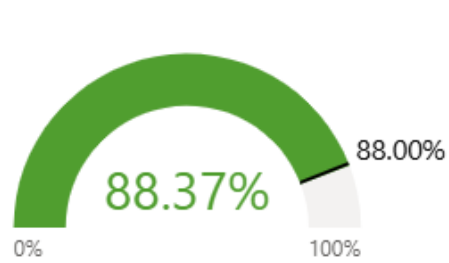
Admit Rate % is the percentage of applicants who are offered admission.

Yield % vs Yield % Benchmark



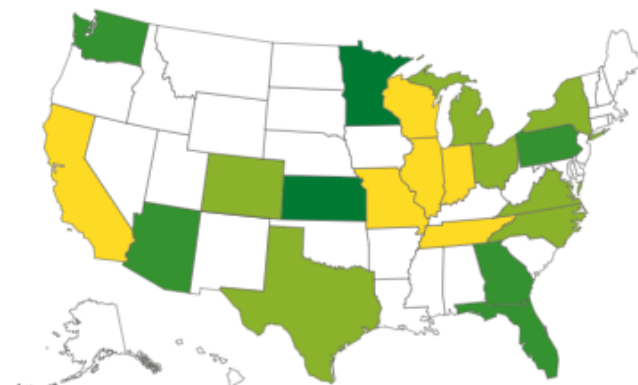
Yield % is the % of admitted students who actually paid a deposit (committed to attend).

A&R % vs A&R % Benchmark

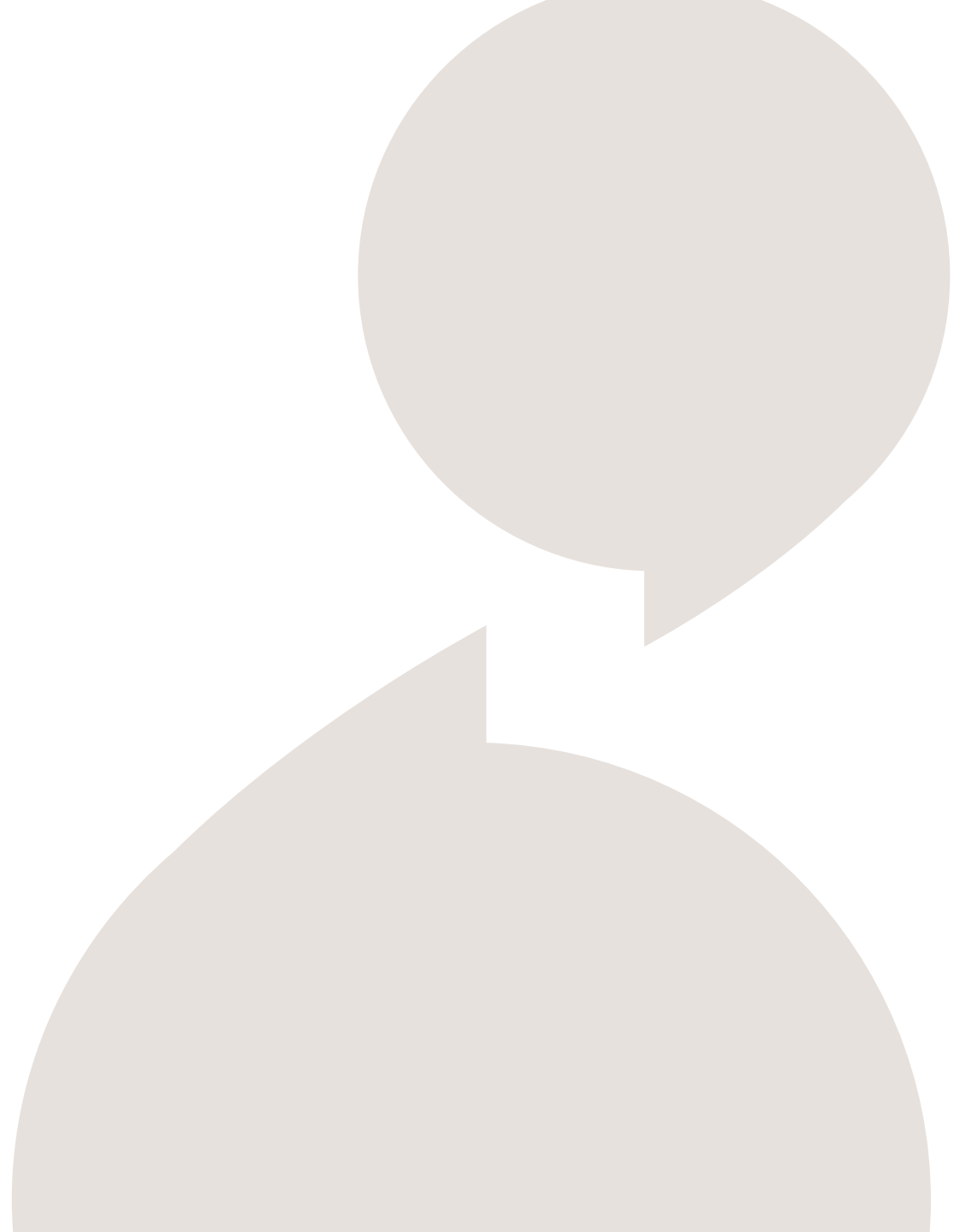


A&R % (Acceptance & Registration Rate) is the % of deposited students who actually show up and enroll.

Net Deposits # by State (Domestic)



Why this works?



The Anatomy of an Effective Dashboard

Great dashboards are intentional.
Every element should help users quickly understand performance and take action.

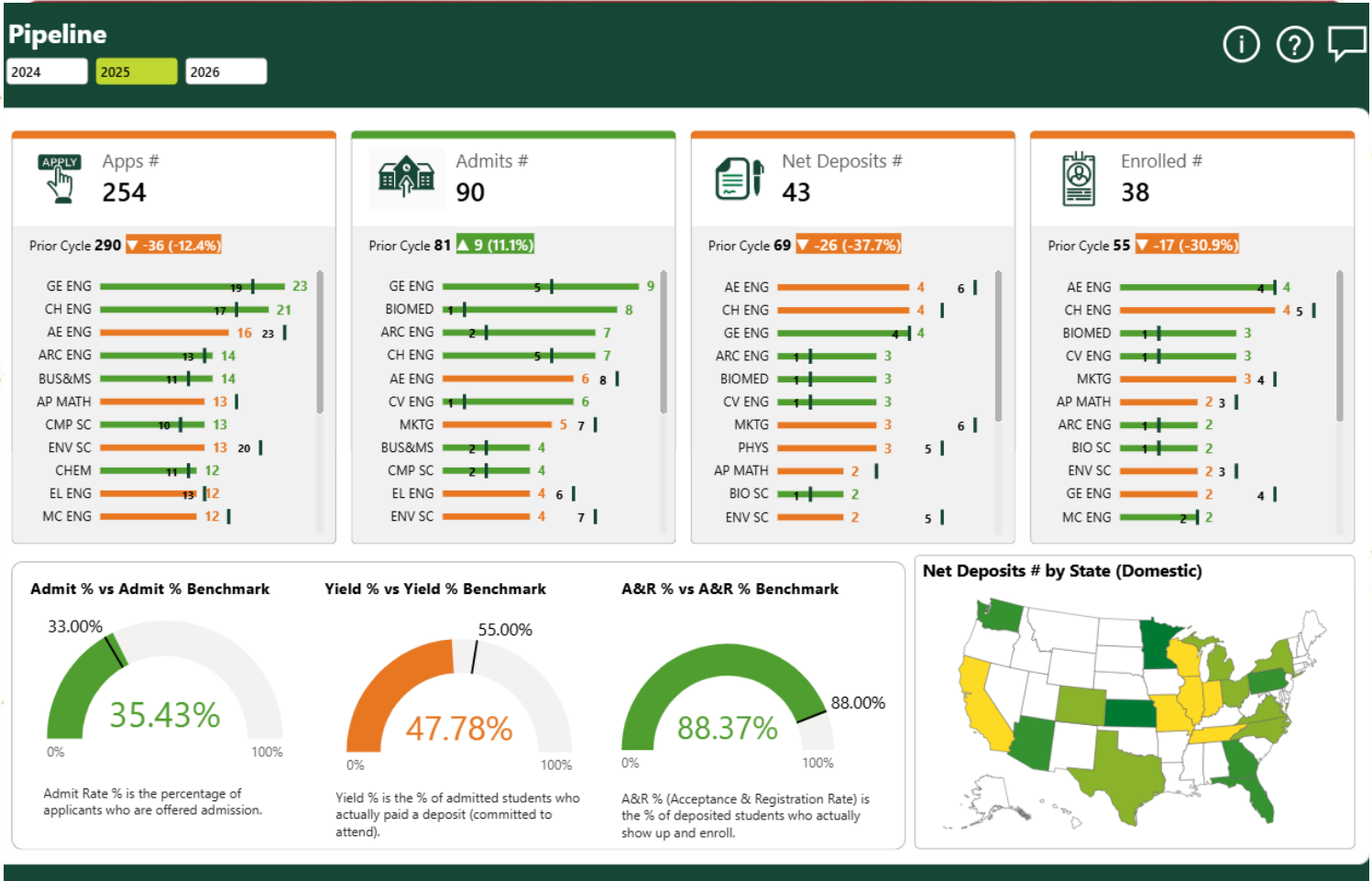
1 Clear KPI Section

2 Comparisons & Benchmarks

3 Visual Hierarchy

4 Consistent Design Language

5 Interactivity with Purpose



Good dashboards don't just show data. They help people see what matters and decide what to do next.

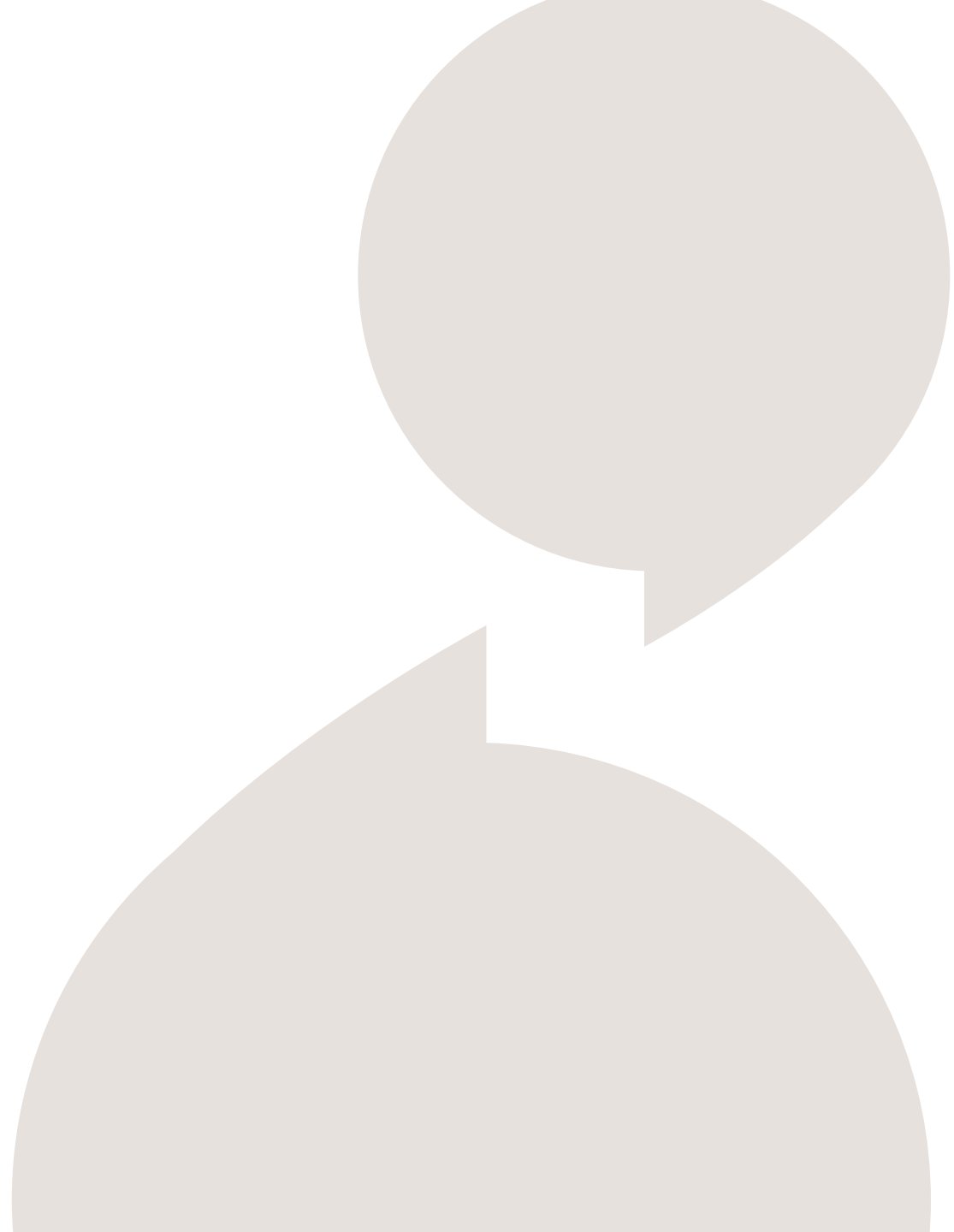
Download resources from

<https://tinyurl.com/AIR2026PowerBI>

Power BI is NOT JUST Drag and Drop



Designing Before Building



Designing Before Building

- 1. Start With Questions, Not Charts**
- 2. Know Your Audience**
3. Define the Core KPIs First
4. Wireframe
5. Storyboard

Before You Start Building, Think About What Matters Most

Great dashboards start with clarity, not complexity.
Answer these 5 critical questions first.



1. PROCESS

What process are we tracking?

Understand the end-to-end process and key steps that drive results.

DASHBOARD EXAMPLE

Enrollment Pipeline funnel:

- 1 Applications
- 2 Admits
- 3 Net Deposits
- 4 Enrolled

WHY IT MATTERS

Helps us see where students are in the funnel and where leakage occurs.



2. OUTCOMES

What outcomes do we care about?

Define the results that indicate success and progress.

DASHBOARD EXAMPLE

Key outcomes displayed at the top:

- Apps #
- Admits #
- Net Deposits #
- Enrolled #

Each with Prior Cycle comparison and % change.

WHY IT MATTERS

Shows performance at a glance and how we're trending vs. prior cycle.



3. USERS

Who will use this dashboard?

Know your audience to tailor the view, language, and detail level.

DASHBOARD EXAMPLE

Designed for enrollment leaders and program/dept. leaders:

- High-level KPIs
- Program (Academic Plan) performance
- State view
- Easy filters to explore

WHY IT MATTERS

The right information, in the right way, for the right people.



4. DECISIONS / ACTIONS

What decisions or actions will this inform?

Dashboards should drive clarity that leads to action.

DASHBOARD EXAMPLE

This dashboard helps leaders:

- Identify programs with low admit or yield rates
- Address weak deposit conversion
- Focus recruitment in strong states
- Allocate resources where impact is highest

WHY IT MATTERS

Turns insights into actions that improve enrollment outcomes.



5. PROBLEMS / QUESTIONS

What problems or questions are we trying to solve?

Start with the questions. The data and visuals should help answer them.

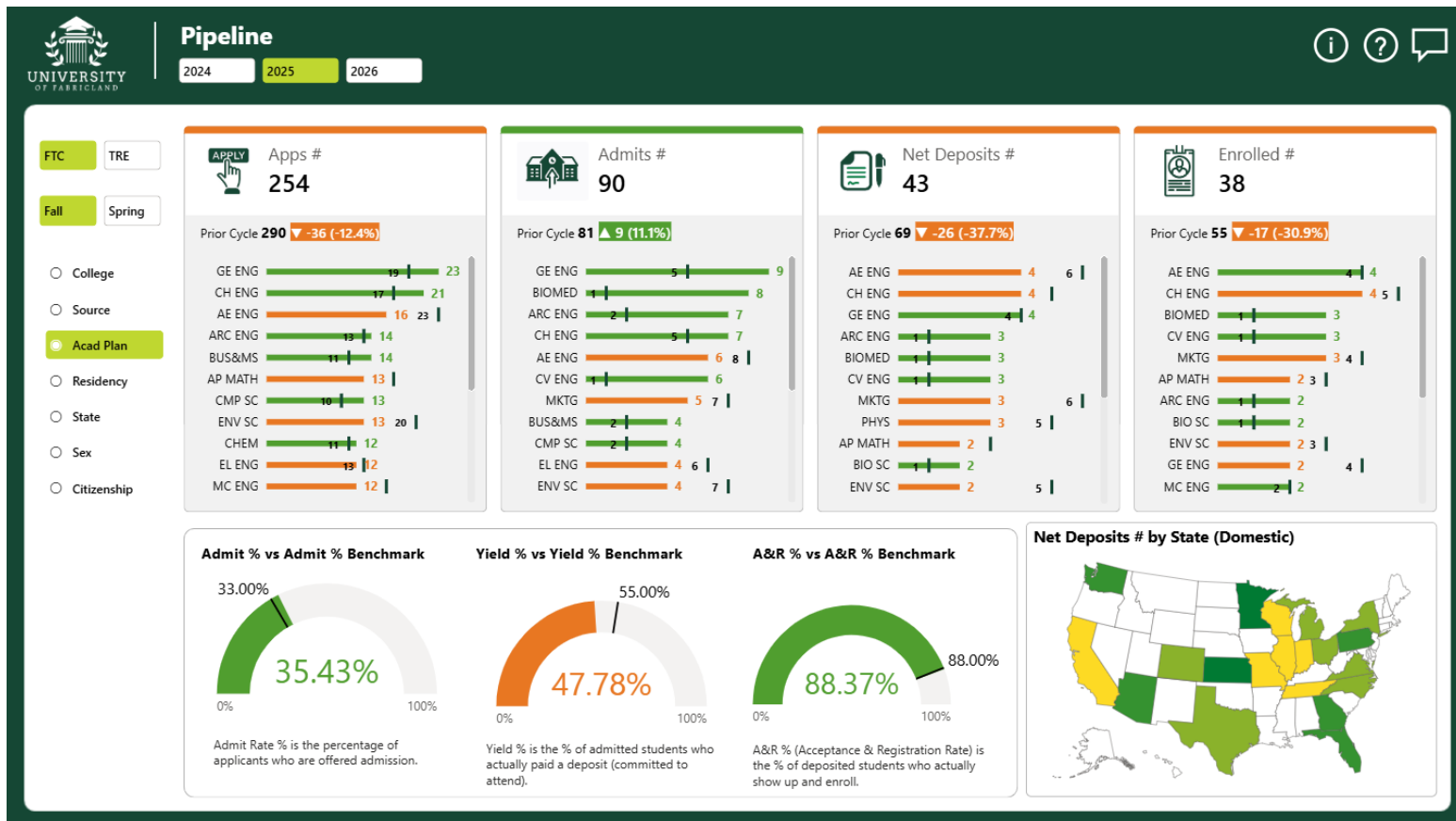
DASHBOARD EXAMPLE

Questions this dashboard helps answer:

- Are applications trending up or down?
- Which programs are underperforming?
- Where is funnel leakage occurring?
- Which states are driving the most deposits?

WHY IT MATTERS

Keeps the dashboard focused on solving real problems.



Recap:

Audience? Purpose?

Designing Before Building

1. Start With Questions, Not Charts
2. Know Your Audience
- 3. Define the Core KPIs First**
4. Wireframe
5. Storyboard

Define the Core KPIs First

KPIs define the story of your dashboard. Get them right before you build.



Why it matters

If you don't know what metrics matter, how they are calculated, and why they matter, the dashboard becomes cluttered and unfocused.



KEY QUESTIONS TO ASK



What are the primary measures?



Which metrics indicate success?



What benchmarks matter?



What comparisons are important?



Which KPIs should users see first?



HOW TO DEFINE YOUR CORE KPIs

1



UNDERSTAND THE GOAL

Clarify the business objectives and decisions the dashboard will support.

2



IDENTIFY KEY OUTCOMES

Define the results that indicate success toward those objectives.

3



SELECT THE RIGHT KPIs

Choose a small set of metrics that best reflect performance and drive decisions.

4



DEFINE & VALIDATE

Ensure each KPI has a clear definition, formula, source, and aligns with user needs.

5



PRIORITIZE VISIBILITY

Decide which KPIs should be surfaced first and which are supporting details.



BEST PRACTICE: PRIORITIZE IN 3 LEVELS

1



MOST IMPORTANT KPIs FIRST

Show the metrics that drive decisions at the top.

2



SUPPORTING DETAILS SECOND

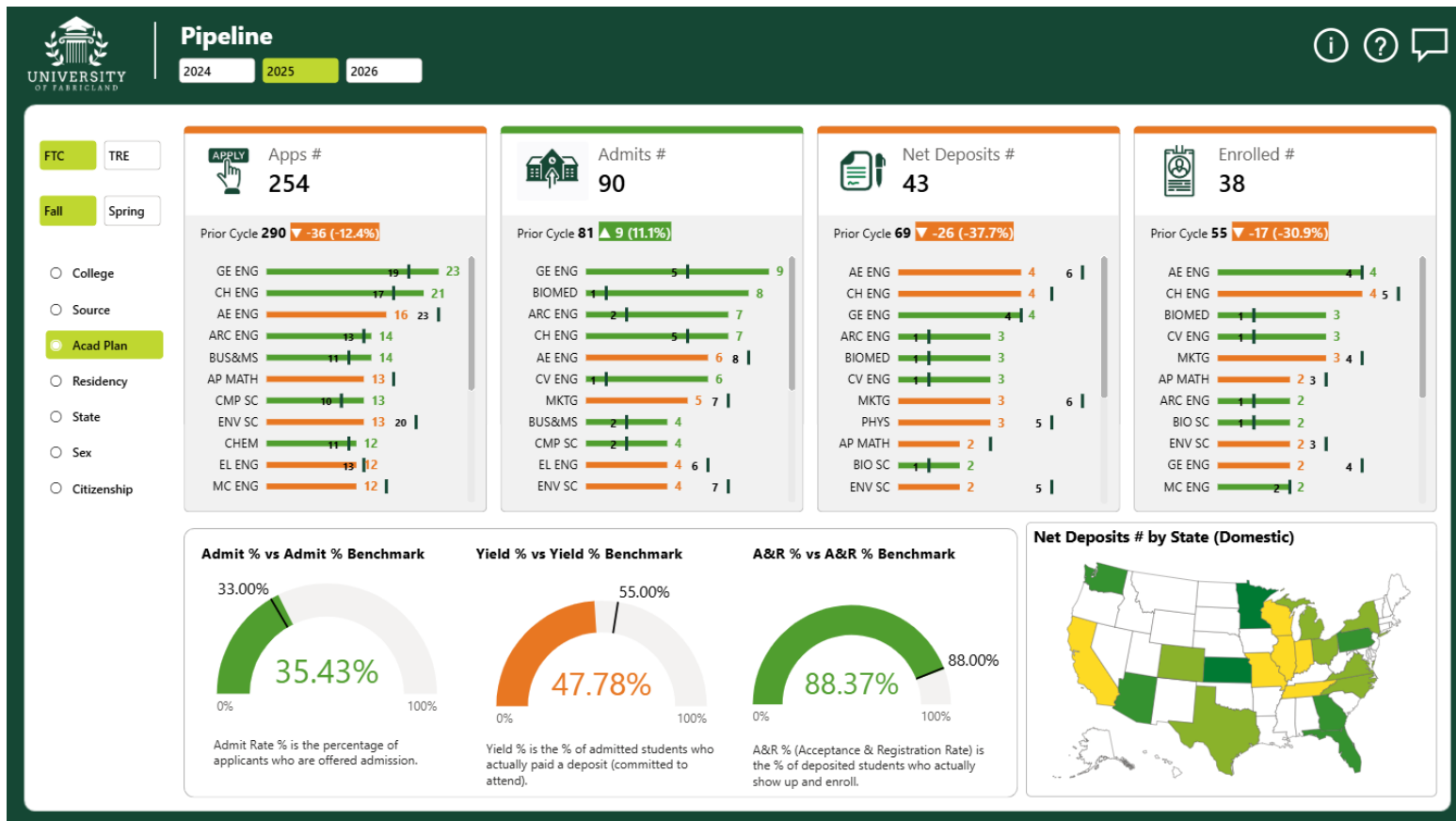
Provide context and breakdowns next.

3



EXPLORATORY ANALYSIS LAST

Allow users to drill down when needed.



Recap:

Key Metrics/KPIs

Designing Before Building

1. Start With Questions, Not Charts
2. Know Your Audience
3. Define the Core KPIs First
- 4. Wireframe**
5. Storyboard

WIREFRAMING YOUR DASHBOARD

Plan the structure first. Build with clarity and purpose.



Why wireframe?

Wireframing helps you define what to show, how it's organized, and how users will experience it before you start building.

WIREFRAMING BEST PRACTICES



Start with the goal
Keep the business questions and decisions in mind.



Identify key sections
Top KPIs, insights, comparisons, breakdowns, and details.



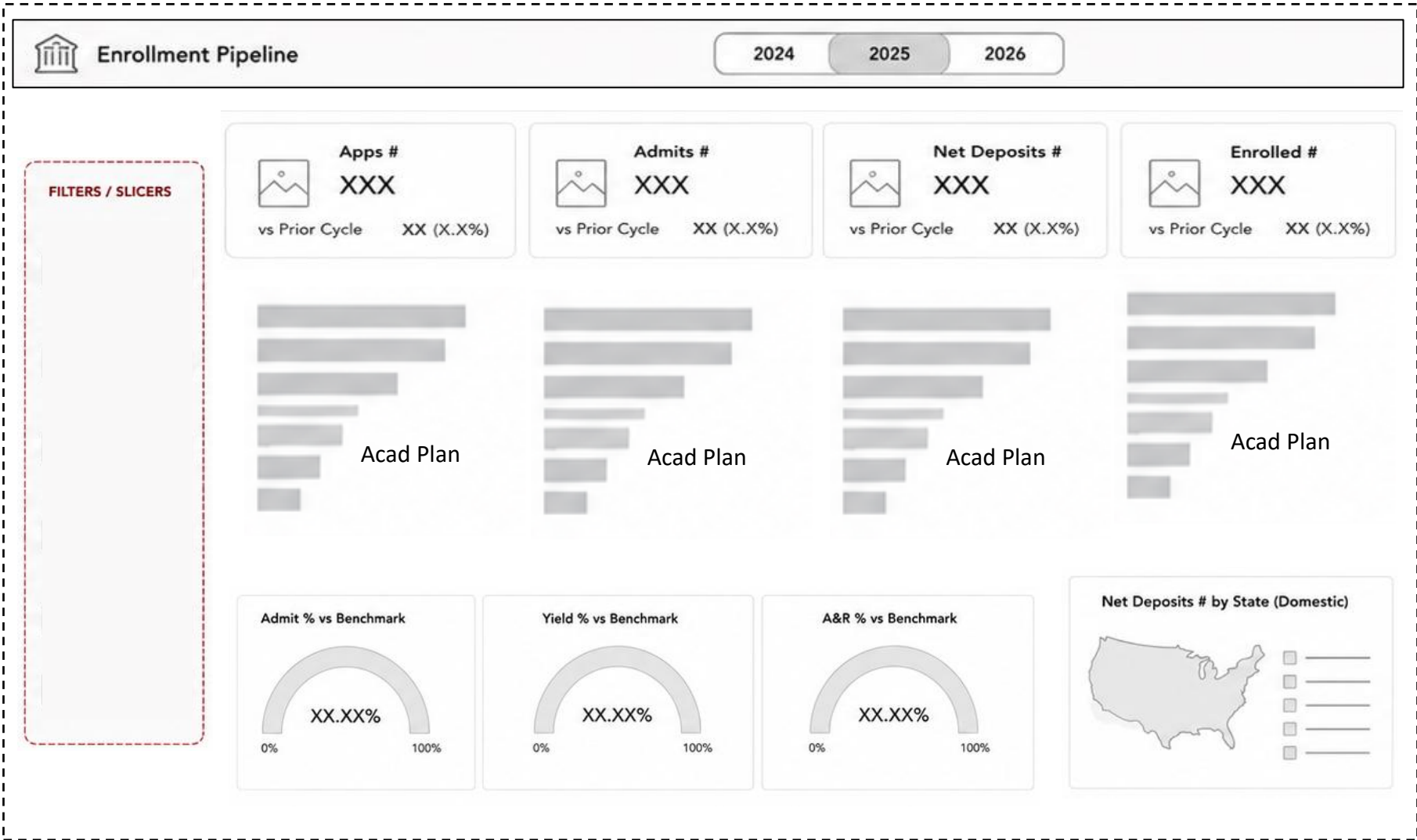
Prioritize information
Place the most important information first.



Keep it simple
Avoid clutter. Show only what drives decisions.



Plan for interactivity
Think about filters, drilldowns, and navigation early.





People don't read dashboards. They scan them.

Design your layout to match natural reading behavior so users can quickly find what matters.

Layout Wireframing in Power BI- DEMO

Designing Before Building

1. Start With Questions, Not Charts
2. Know Your Audience
3. Define the Core KPIs First
4. Wireframe
- 5. Storyboard**

STORYBOARDING YOUR DASHBOARD

Map the story before you build. A clear story creates a clear dashboard.



Why storyboard?

Storyboarding helps you define the flow of insights, the key messages on each page, and how users will move through the data.

1

STARTING POINT – INITIAL WIREFRAME

Initial wireframe focused only on Academic Plan.



STORYBOARDING YOUR DASHBOARD

Map the story before you build. A clear story creates a clear dashboard.



Why storyboard?

Storyboarding helps you define the flow of insights, the key messages on each page, and how users will move through the data.

2 ANALYSIS – IDENTIFYING THE GAP

Reviewing the data, we noticed large differences across other segments.



Limiting to Acad Plan hides important variation across other segments

STORYBOARDING YOUR DASHBOARD

Map the story before you build. A clear story creates a clear dashboard.



Why storyboard?

Storyboarding helps you define the flow of insights, the key messages on each page, and how users will move through the data.

3 DECISION – EXPAND TO KEY SEGMENTS

Decided to include multiple segments for a fuller, multi-dimensional view.

Showing other key segments by using Field Parameters in Power BI



STORYBOARDING YOUR DASHBOARD

Map the story before you build. A clear story creates a clear dashboard.



Why storyboard?

Storyboarding helps you define the flow of insights, the key messages on each page, and how users will move through the data.

4 GEOGRAPHIC INSIGHT – NEED MORE CONTEXT

The map is useful, but users need more details by state.



STORYBOARDING YOUR DASHBOARD

Map the story before you build. A clear story creates a clear dashboard.



Why storyboard?

Storyboarding helps you define the flow of insights, the key messages on each page, and how users will move through the data.

5 SOLUTION – ENHANCED TOOLTIP FOR MAP

Add rich tooltip to the map to provide deeper insights on demand.



Arizona

Net Deposits #

419

Prior Cycle 9

Variance ▲ 410 (4555.6%)

CHECKLIST: WIREFRAMING & STORYBOARDING

Use these checklists to plan with purpose and design dashboards that tell the right story and drive action.



WIREFRAMING CHECKLIST

Purpose: Define the structure and layout before building.



1. PURPOSE & GOALS

What problem are we solving?

- ☐ What decisions will this support?
- ☐ What questions must it answer?
- ☐ What actions should users take?
- ☐ Is the dashboard focused on one clear purpose?



2. AUDIENCE

Who are the primary users?

- ☐ What level of detail do they need?
- ☐ What KPIs matter most to them?
- ☐ How familiar are they with data?
- ☐ Where will this be viewed?



3. KPIs & METRICS

Have the core KPIs been identified?

- ☐ Are KPI definitions agreed?
- ☐ Are benchmarks/comparisons needed?
- ☐ Are trends/prior comparisons included?
- ☐ Are most important KPIs prioritized?



4. LAYOUT & STRUCTURE

Is the layout easy to scan?

- ☐ Does it follow a logical reading pattern?
- ☐ Are visuals grouped logically?
- ☐ Is there a clear visual hierarchy?
- ☐ Are high-priority items on top?



5. VISUAL PLANNING

Is the right chart type chosen?

- ☐ Does each visual answer a question?
- ☐ Are there too many visuals?
- ☐ Is whitespace used effectively?
- ☐ Is clutter minimized?



6. FILTERS & INTERACTIVITY

Which slicers are truly necessary?

- ☐ Are filters intuitive & easy to find?
- ☐ Will users need drillthrough/tooltip?
- ☐ Are navigation paths clear?
- ☐ Does interactivity simplify the experience?



7. ACCESSIBILITY

- ☐ Is color contrast readable?
- ☐ Are colors used consistently?
- ☐ Can it be understood without relying only on color?
- ☐ Are font sizes readable?
- ☐ Are labels and titles meaningful?



8. FINAL VALIDATION

- ☐ Can users find important insights within seconds?
- ☐ Does the layout feel balanced?
- ☐ Is anything unnecessary?
- ☐ Does the design support fast decision-making?
- ☐ Would a first-time user understand it quickly?



STORYBOARDING CHECKLIST

Purpose: Define the flow of insights and the user's journey.



1. STORY & NARRATIVE

- ☐ What is the main story?
- ☐ What should users understand first?
- ☐ What insights come next?
- ☐ Is the narrative flow intentional?
- ☐ Does it guide users naturally?



2. USER JOURNEY

- ☐ What path will users take?
- ☐ Overview → Analysis → Action?
- ☐ Does it reduce confusion?
- ☐ Are key insights easy to locate?
- ☐ Does each section have a purpose?



3. INSIGHT VALIDATION

- ☐ Does each visual contribute to the story?
- ☐ Are insights actionable?
- ☐ Are comparisons meaningful?
- ☐ Are trends & anomalies highlighted?
- ☐ Are supporting details available?



4. DECISION SUPPORT

- ☐ What decisions can users make?
- ☐ Are risks clearly visible?
- ☐ Are opportunities highlighted?
- ☐ Are important changes easy to identify?
- ☐ Does it encourage action?



5. DRILLDOWNS & DEEPER INSIGHTS

- ☐ Do users need more context?
- ☐ Should tooltips provide insights?
- ☐ Are drillthrough paths meaningful?
- ☐ Can users explore segments easily?
- ☐ Is detailed info available on demand?



6. STORY FLOW VALIDATION

- ☐ Does it answer the intended questions?
- ☐ Is information in the right order?
- ☐ Does the experience feel intuitive?
- ☐ Is it focused, not overwhelming?
- ☐ Would leadership quickly get the key takeaways?



REMEMBER

A great dashboard is not about more data, it's about the right insights in the right order.



Wireframing defines the structure.

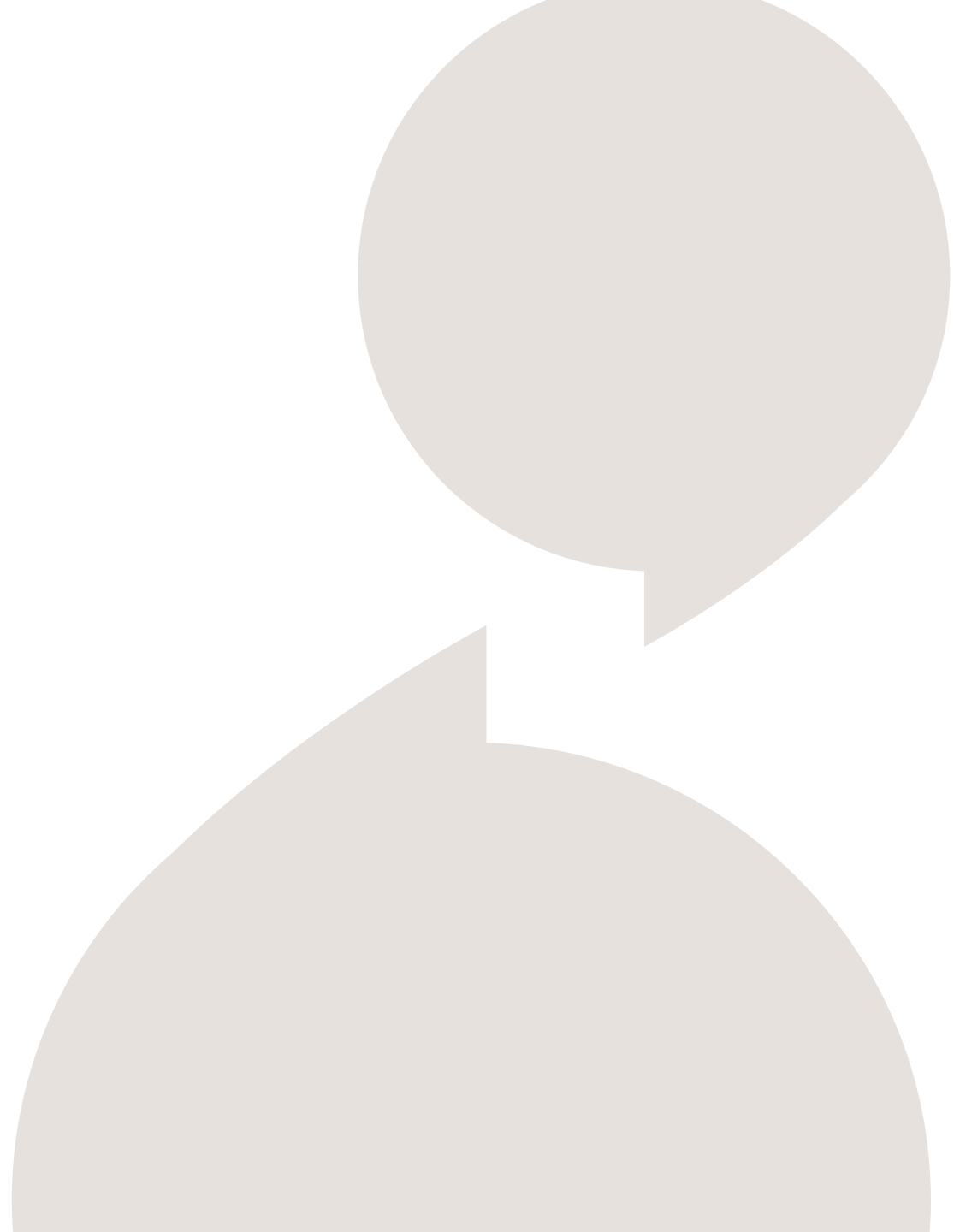


Storyboarding defines the experience.



Together, they turn data into insights and insights into action.

Designing



KISS PRINCIPLE

KEEP IT SIMPLE, STUPID



FOCUS ON THE GOAL

Start with a clear objective. Show only what helps answer the key question.



SIMPLIFY

Remove anything that doesn't add value. Eliminate clutter, unnecessary visuals, and duplicate information.



USE THE RIGHT VISUALS

Choose clear, familiar charts that best communicate the message instantly.



HIGHLIGHT WHAT MATTERS

Use hierarchy, spacing, and emphasis to draw attention to key insights.



KEEP IT CONSISTENT

Use consistent colors, fonts, labels, and layouts for a smooth user experience.



MAKE IT EASY TO ACT

Ensure users can quickly understand the insights and know what action to take.

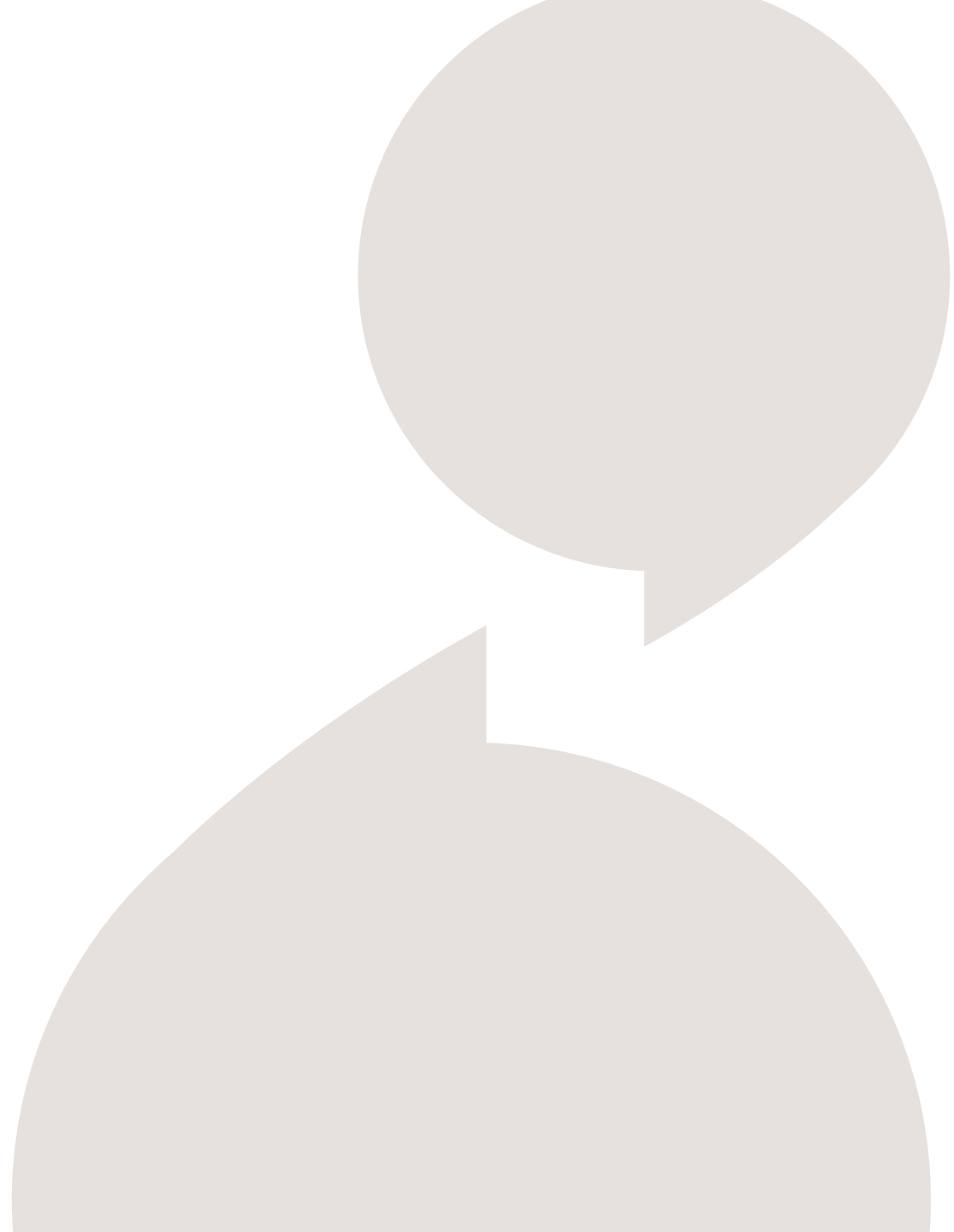


Simplicity is not about showing less data, but showing the **right data clearly**.

Why it matters

Simple dashboards are easier to understand, faster to interpret, and drive better decisions.

Hands-on



Game Plan



Part 1 : Background and Theme

Part 2 : KPI Cards

Part 3 : Benchmark Visuals

Part 4 : Map Visual

Part 5 : Final Polish

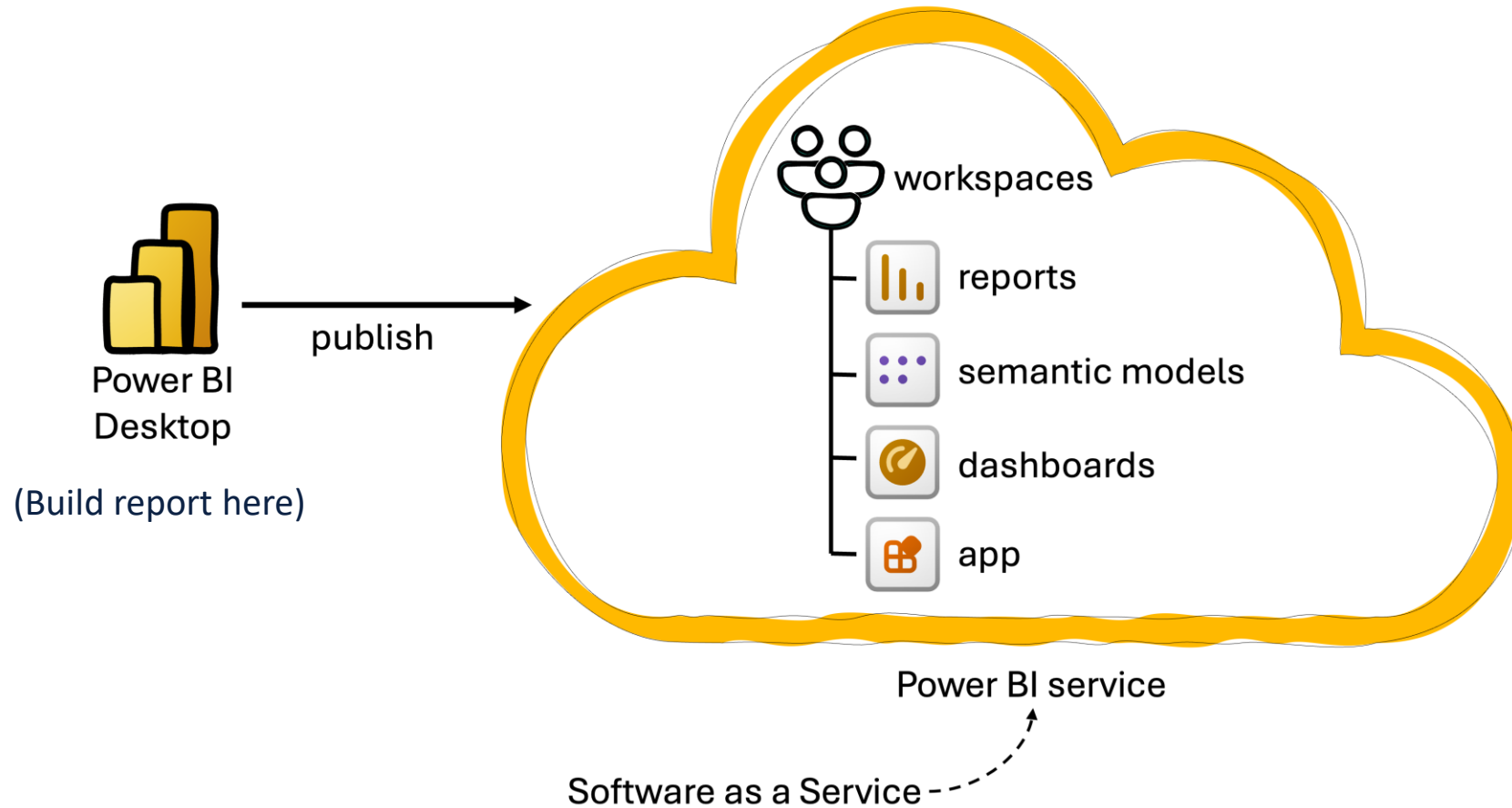
Part 6 : BONUS- SVG's, Native Visual Enhancements

Hands-on Exercise

Download resources from

<https://tinyurl.com/AIR2026PowerBI>

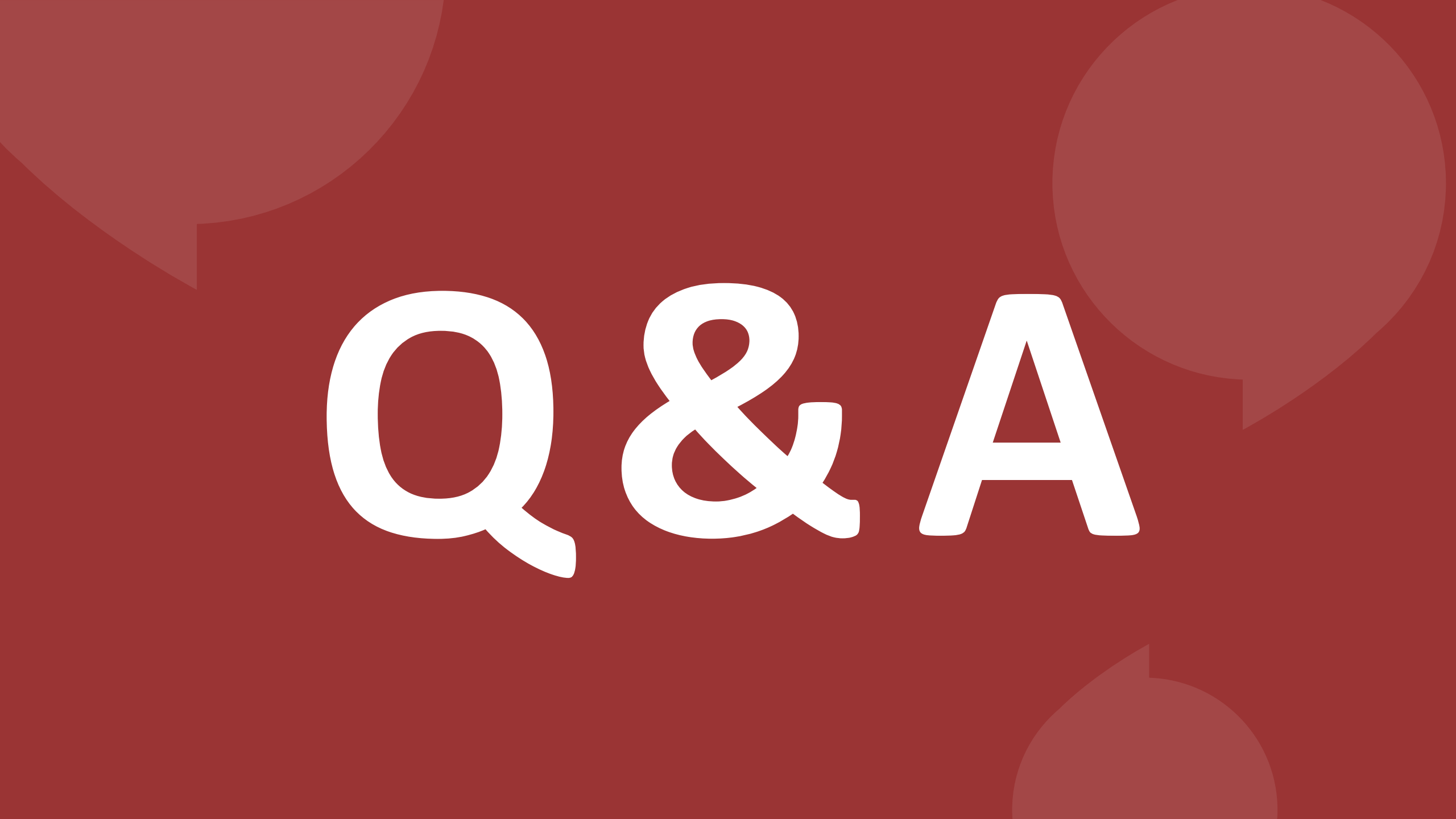
Power BI Service



It's the Software as a Service (SaaS) part of Power BI and can be accessed through a web browser. <https://app.powerbi.com>



**Don't forget to rate our session
and leave your feedback!**

The background is a solid dark red color. There are three white speech bubble shapes scattered across the image: one in the top left, one in the top right, and one in the bottom right. The text "Q & A" is centered in the middle of the image in a white, bold, sans-serif font.

Q & A